OPERATION: FIESTA NIGHT RESCUE

NET 301 — Chapter 6 Quality of Service
Network Rescue Report

QoS Rescue Team
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Section: 3-IT11

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Prepared for Missions 0–5, Twist Response, Mission 6 simulation preparation, and Council Hearing.

Executive Summary

Barangay Sigla's Fiesta Hub has a 100 Mbps LAN serving as an input to a 2.048 Mbps (2048 kbps) WAN link. Hence, the peak traffic demand of 2,836 kbps exceeds the capacity of the WAN link by 788 kbps. The project team correctly identified that the cause of congestion is the LAN-to-WAN speed mismatch and traffic aggregation (Slide 8). Therefore, the proposed voice solution uses LLQ, while the video, payments, staff web/social traffic, and sales backup employ CBWFQ.

Mission 0 was updated with the newly supplied morning and afternoon Speedtest evidence, plus the formerly recorded evening evidence. If a metric is not present in a screenshot, this fact is explicitly stated as unavailable instead of being invented.

Mission 0 — Field Recon (10 pts)

Location: same home Wi-Fi test location. Device count recorded previously for evening: 10 connected devices.

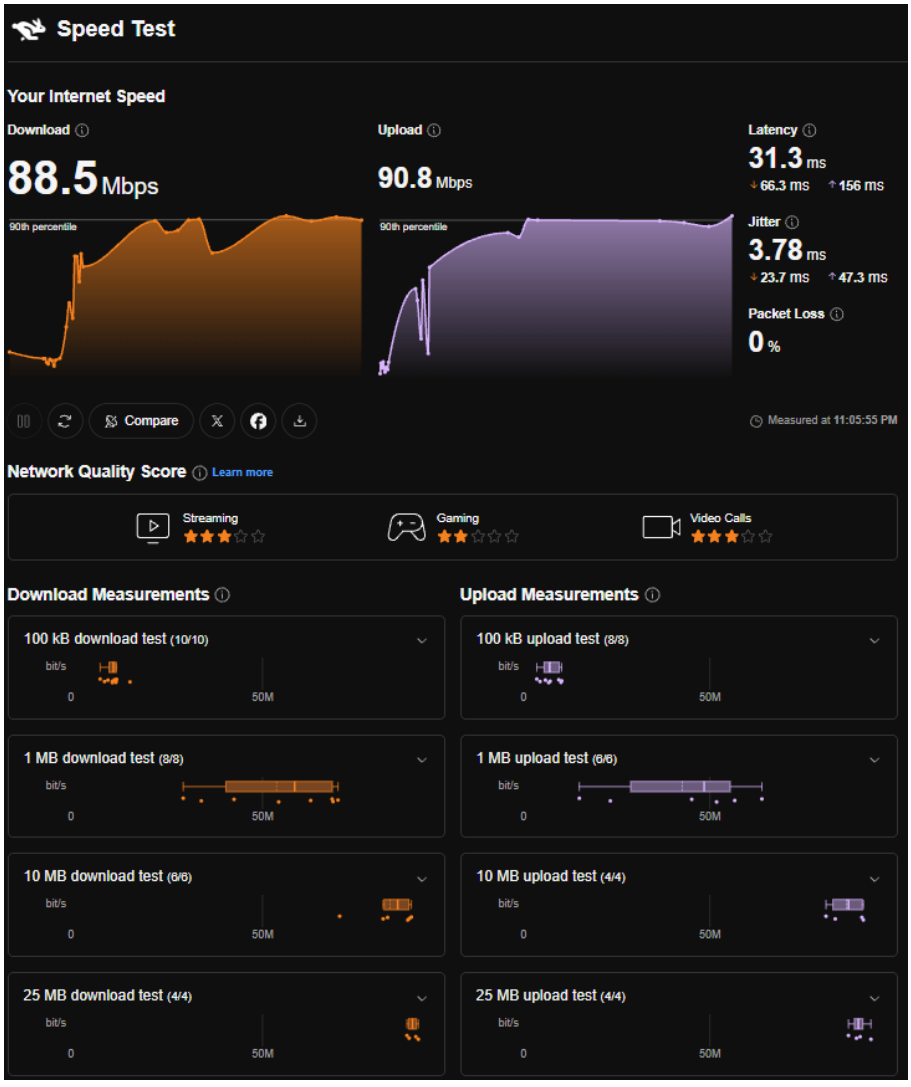


Figure 1. Morning Speedtest — 88.5 Mbps download, 31.3 ms latency, 3.78 ms jitter, 0% packet loss.

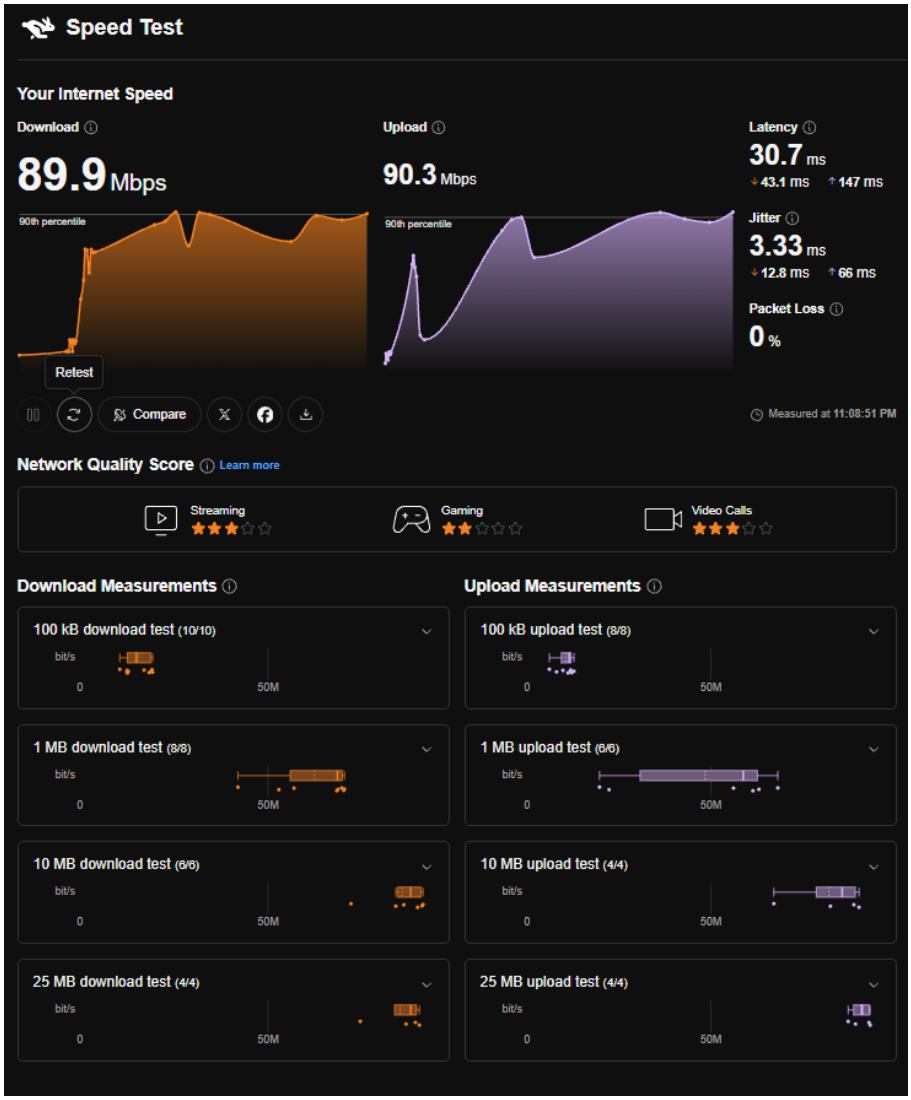


Figure 2. Afternoon Speedtest — 89.9 Mbps download, 30.7 ms latency, 3.33 ms jitter, 0% packet loss.

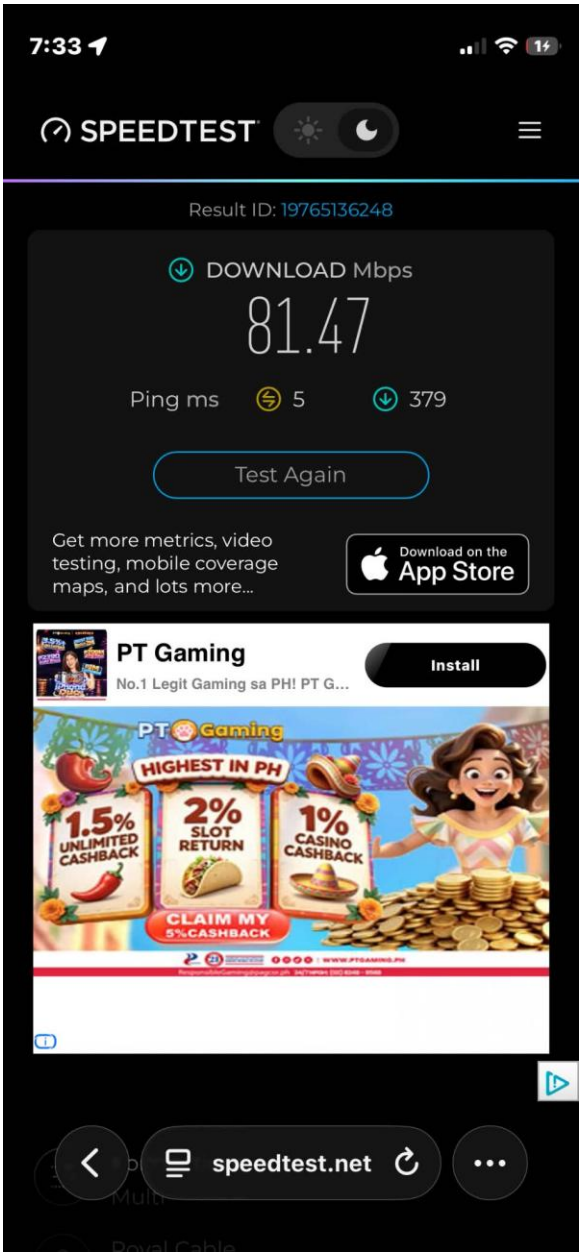


Figure 3. Evening Speedtest — 81.47 Mbps download and 5 ms ping. This screenshot does not display jitter or packet loss.

Time	Download	Latency/Ping	Jitter	Packet Loss	Assessment
Morning	88.5 Mbps	31.3 ms	3.78 ms	0%	Pass on shown metrics
Afternoon	89.9 Mbps	30.7 ms	3.33 ms	0%	Pass on shown metrics
Evening peak	81.47 Mbps	5 ms	Not shown	Not shown	Download/ping shown; jitter/loss not available

Comparison: The morning and afternoon readings show stable, high throughput with low latency/jitter and no packet loss. The evening screenshot shows a lower download rate than the daytime tests, but still 81.47 Mbps. Since the evening image does not show jitter/loss, these values are not being claimed.

Congestion point: LAN-to-WAN speed mismatch plus aggregation. The home test itself is not the Sigla Hub, but it shows that the internet connection could remain 'working' while the quality depends on the traffic load and timing (Slide 8).

Video-call observation: During the 5-minute call, the video was not blurry, jagged, or frozen; audio did not cut out and was not noticeably out of sync. There was a little delay, however. This is in line with some minimal amount of delay without strong evidence of packet-loss-induced voice/video failure (Slides 9–10, 14).

Mission 1 — Traffic Detective (15 pts)

Part A — Traffic Profile

Traffic	TCP/UDP	Drop-sensitive?	Delay-sensitive?	Interactive?	Mission-critical?	Priority & reason
Voice (VoIP)	UDP	Yes	Yes	Yes	Yes	High. real-time voice is highly sensitive to delay/jitter/loss (Slide 13).
Video	UDP	Yes	Yes	Yes	No	High/Medium. real-time video is delay/jitter sensitive (Slide 14).
Payments	TCP	Yes	Yes	Yes	Yes	High. transaction confirmation is mission-critical.
Sales backup (FTP)	TCP	No/less	No/less	No	No	Low. important to owner, but not interactive; can be delayed.
Staff web/social	TCP	Moderate	Moderate	Yes	No	Medium. interactive but not mission-critical.

Part B — Calculations

1) Total demand = 768 + 768 + 200 + 800 + 300 = 2,836 kbps.

2) WAN capacity = 2,048 kbps. Overload = 2,836 − 2,048 = 788 kbps. Demand as a percentage of capacity = (2,836 ÷ 2,048) × 100 = 138.48%. Therefore, demand is 38.48% above available capacity.

3) Voice + video + payments = 768 + 768 + 200 = 1,736 kbps. Percentage of link = (1,736 ÷ 2,048) × 100 = 84.77%. Remaining = 2,048 − 1,736 = 312 kbps. Web + backup demand = 300 + 800 = 1,100 kbps. Shortfall = 1,100 − 312 = 788 kbps.

Part C — Logs

Traffic	Latency	Jitter	Loss	Result
Voice	210 ms — FAIL	45 ms — FAIL	3% — FAIL	Fails all three referenced voice limits (Slide 13).
Live-stream video	320 ms — PASS within stated range	60 ms — FAIL	0.5% — PASS	Jitter exceeds the cited course range; latency/loss are acceptable (Slide 14).

Playout delay buffer: temporarily store voice packets so that packets arriving at different times can be played out at a more steady rate. Excessive jitter can cause packets to arrive too late in the useful reception window, creating gaps/dropouts (Slide 9).

Congestion related delay that builds as the link saturates is queuing delay (Slide 8). Fixed delays such as processing, serialization/packetization, and propagation components are not simply avoided through QoS; QoS primarily focuses on managing variable components such as queuing/congestion behavior.

Root cause at the Sigla Hub: aggregation of several classes of traffic into a 100 Mbps LAN-to-2.048 Mbps WAN bottleneck, i.e., a LAN-to-WAN speed mismatch plus aggregation (Slide 8).

Mission 2 — Queue Engineer (20 pts)

Part A — Algorithm Selection

Situation	Choice	Why
1. One PC, fast link, almost never busy	FIFO	Simple and sufficient if congestion unlikely; no complex classification needed.
2. Automatic fair sharing among many conversations, interactive traffic moved forward, no manual classes	WFQ	Provides automatic fair sharing of bandwidth with differentiated treatment without having to manually define classes.
3. Admin defines classes and guarantees bandwidth	CBWFQ	Each class can be explicitly defined and allocated a guaranteed amount of bandwidth during congestion..
4. Voice needs strict priority and minimal jitter	LLQ	LLQ provides a strict priority queue for delay/jitter sensitive voice traffic..
5. Current setup with payment waiting behind FTP	FIFO	FIFO effectively serves packets based on arrival order, so a large non-critical flow could potentially delay a critical packet.
6. Sigla WAN	LLQ + CBWFQ	LLQ protects voice, and CBWFQ manages the other classes and makes explicit trade-offs

Part B — Bandwidth Policy

Class	Queue	Guaranteed bandwidth	Why
Voice	LLQ priority	768 kbps	Exactly carries 12 × 64 kbps calls; protects real-time voice.
Video	CBWFQ	768 kbps	Matches 2 × 384 kbps video demand.
Payments	CBWFQ	200 kbps	Matches mission-critical transaction demand.
Staff web	CBWFQ	100 kbps	Gets service but accepts a 200 kbps peak shortfall.
Sales backup (FTP)	CBWFQ	212 kbps	Receives guaranteed service while accepting a 588 kbps shortfall during peak congestion.
TOTAL		2048 kbps	Exactly equals WAN capacity.

Trade-off: voice, video, and payments consume 1,736 kbps, leaving 312 kbps for web and FTP. The policy therefore sacrifices lower-priority peak throughput so that critical real-time/transaction traffic remains usable.

Part C — LLQ Diagram

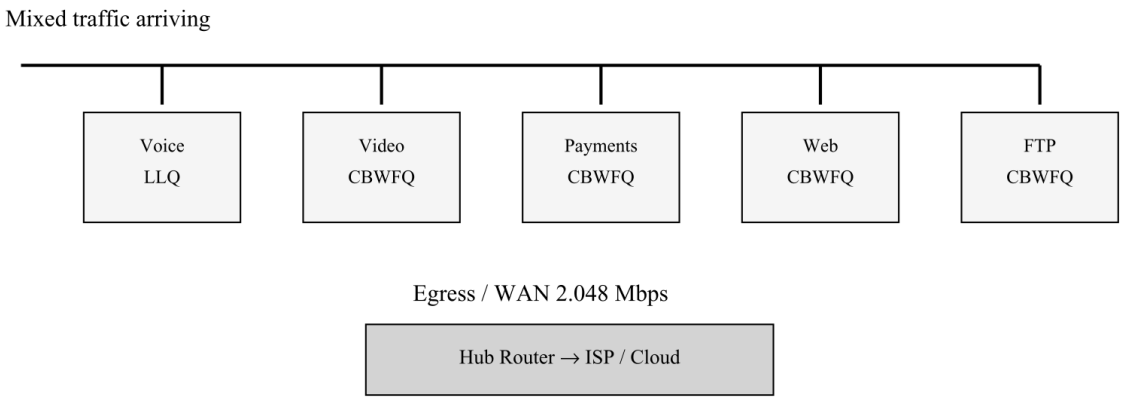


Figure 4. LLQ + CBWFQ policy: voice is strict priority; other classes use guaranteed CBWFQ service (Slide 21).

Mission 3 — Model Pitch (10 pts)

Model	How it works	Benefits	Drawbacks	Scalability	Guaranteed delivery?
Best-Effort	Packets are forwarded without reserved per-flow resources.	Simple, low overhead, scalable.	No QoS guarantee; delay/loss can vary.	High	No
IntServ	Per-flow signaling/reservation provides resources end-to-end.	Strong per-flow guarantees; admission control.	Resource/signaling heavy; state per flow.	Lower	Can provide strong guarantees if resources are available
DiffServ	Classifies/marks packets and applies class-based treatment hop-by-hop.	Scalable, practical for many flows/routers.	Requires consistent classification, marking, and policy.	High	No absolute per-flow guarantee

Part B — Situation Matching

Situation	Best model	Reason
1. Small sari-sari store free Wi-Fi	Best-Effort	Simple service where strict guarantees are unnecessary.
2. Telemedicine with few guaranteed flows	IntServ	Per-flow reservation/admission control fits the need to protect existing calls.
3. Many branches and routers	DiffServ	Scalable class-based hop-by-hop treatment fits a large network.

Part C — Recommendation & Filipino Analogies

Recommendation: DiffServ is the best overall QoS model for TindaTayo Express because the company may have many devices, flows, and routers. IntServ is useful for special environments requiring per-flow guarantees, but deploying per-flow reservations everywhere is less scalable (Slide 24).

Model	Filipino analogy
Best-Effort	Regular jeepney queue: boarding in regular order without a reserved seat or lane
IntServ	Reserved tricycle/vehicle: a specific passenger gets a specific resource; new requests can be denied when the resource is unavailable

DiffServ	Fiesta traffic system with priority lanes: traffic is grouped into classes; each class is treated differently

Mission 4 — Marking Specialist (15 pts)

Traffic	Layer 2 CoS	Layer 3 DSCP name	DSCP decimal	Logic
Voice	5	EF	46	Cisco-recommended EF marking for voice; highest priority.
Video	4	AF41	34	High-priority assured forwarding for real-time video.
Payments	3	AF31	26	Important transaction traffic gets a high assured-forwarding class.
Staff web	1	AF21	18	Medium/lower class for interactive but noncritical traffic.
Sales backup	0	AF11	10	Lower assured-forwarding class for noninteractive bulk traffic.
Everything else	0	BE	0	Best effort/default.

Marking logic: The ranking assigns strict priority to voice, high treatment for video and payments, and low for web/backup. Mark the traffic as close to the source as possible, in order that classification represents actual application behavior on traffic crossing the network (Slides 34-36).

Part B — Decode the Bits

Item	6-bit DSCP	Decimal
AF43	100110	38
AF21	010010	18
AF13	001110	14
DSCP 26	011010	AF31
DSCP 30	011110	AF33

Correction note: the bit values above follow the standard AF codepoint structure. The AF32 example in the project is 011 10 0 = 28 (Slide 36).

802.1p CoS uses 3 bits, giving $2^3 = 8$ priority values (0–7). DSCP uses 6 bits, giving $2^6 = 64$ codepoints (0–63).

Part C — Trust Boundary

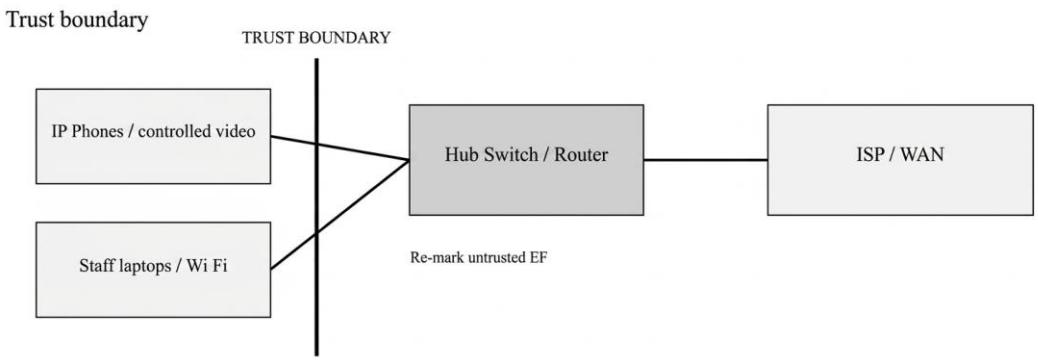


Figure 5. Trust boundary sketch. Controlled IP phones/video endpoints may be trusted; staff laptops not so much. Re-mark untrusted traffic at the boundary.

Traffic should be marked close to the source, so the network can recognize/apply consistent classification early. If your legacy switch isn't IP-aware, you could use Layer 2 CoS on the local segment, but DSCP is preferred across routed L3 networks (Slide 33).

Sneaky Staff Case: If staff laptop marks FTP as EF, and the network trusts that marking, then FTP traffic would enter the strict-priority voice queue and consume capacity. This could cause increased voice delay/jitter or voice drops. Trust boundary should reject/re-mark unauthorized EF markings, and only trust controlled (phone) endpoints.

Mission 5 — Stop the Flood (10 pts)

Part A — WRED

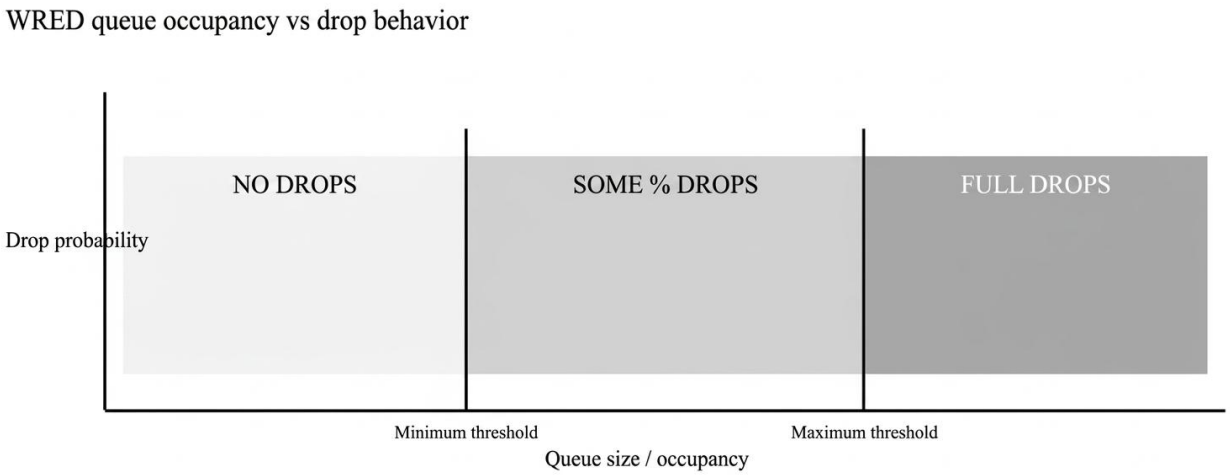
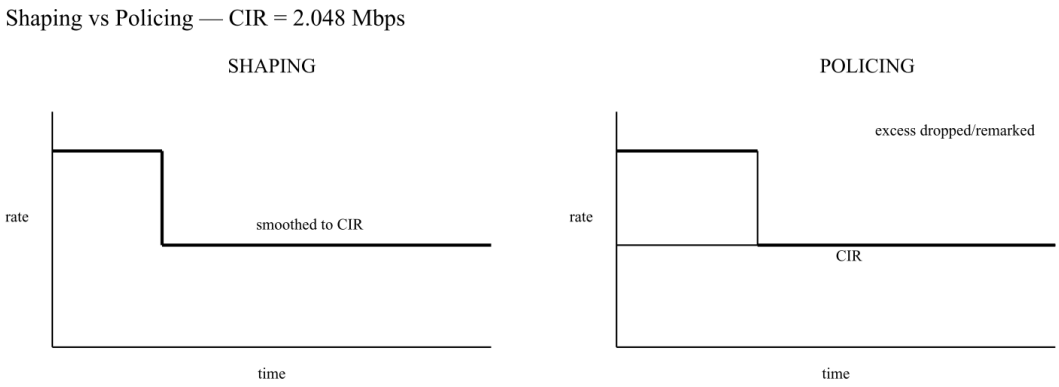


Figure 6. WRED zones: below minimum threshold no early drops; between thresholds increasing early drops; above maximum threshold full drops (Slides 31, 38, 39).

WRED differs from tail drop in that it begins dropping/marking selected packets before the queue is completely full, thus allowing TCP sources to infer congestion and reduce window sizes accordingly. Tail drop occurs when a queue is full, leading to subsequent packets being dropped.

WRED is not the preferred mechanism for providing congestion avoidance for UDP-based voice traffic. Voice requires strict priority and guaranteed service via LLQ; congestion avoidance is only useful for elastic traffic such as TCP.Part B — Shaping vs Policing



Item	Shaping	Policing
Excess packets	Queued/smoothed	Dropped or remarked
Memory	Requires a buffer/queue	Usually does not queue excess for long-term smoothing
Delay	Can add delay because packets wait in queue	Does not intentionally add queueing delay to excess traffic; excess is dropped/remarked
Typical direction	Outbound	Often inbound or at an enforcement point
Sigla example	Hub shapes outbound backup toward 2.048 Mbps	ISP polices traffic above CIR

The ISP owns/enforces the CIR, so the ISP performs policing. The hub should shape its outbound backup traffic before it reaches the ISP so bursts are smoothed to the 2.048 Mbps contract rate. The backup server can burst up to 10 Mbps, so shaping prevents the burst from overwhelming the WAN/ISP handoff (Slide 40).

Twist Response

Twist A — 17 Simultaneous Calls

17 voice calls × 64 kbps = 1,088 kbps. Add video (768 kbps) and payments (200 kbps): 1,088 + 768 + 200 = 2,056 kbps.

Compared with 2,048 kbps WAN capacity: 2,056 − 2,048 = 8 kbps over. Therefore, voice + video + payments do NOT fit; they exceed the link by 8 kbps.

Solution	Effect
1. Admission control / reject the 17th call	Keeps existing calls protected; aligns with IntServ-style admission control (Slide 26).
2. Reduce video bitrate	Frees WAN capacity but reduces video quality.
3. Increase WAN capacity	Solves the capacity problem but costs more and may not be immediately available.

Best choice: admission control/reject the 17th call, because protecting existing calls is better than allowing a new call to degrade all active voice/video/payment traffic.

Twist B – ISP Warning

If the line speed drops below the CIR, the hub should shape outbound traffic to 2.048 Mbps before the ISP. Backup traffic must be queued and smoothed, it must not be allowed to burst above the contracted rate. The ISP is the policing point.

Twist C – Rider App Marks Everything EF

The danger is that if the router trusted the rider app's EF marking, it would allow all the rider traffic into the strict-priority queue, where it would fight with the voice traffic for space, potentially starving the voice traffic. The trust boundary must classify the traffic coming from trusted endpoints and re-mark any unauthorized EF markings. The EF/LLQ classification must be used only for legitimate voice traffic.

Mission 6 – Packet Jeepney Simulation (10 pts)

For the actual physical simulation, we used 20 packet slips: Voice ×6, Video ×4, Payments ×3, Web ×4, and FTP ×3. The router buffer can hold 14 packets, the number of packets sent in the WAN every tick or period was 4 packets, and we used the same packet order for both rounds.

Actual packet order: F3, VID3, W1, F1, P3, VID1, V1, P2, P1, VID2, V2, V3, V5, V6, V4, VID4, W2, W3, W4, F2.

Round 1 – FIFO

We put the first 14 packets in the buffer. The six tail-dropped packets were V4, VID4, W2, W3, W4, and F2.

Voice average = (2 + 3 + 3 + 4 + 4) ÷ 5 = 3.20. The dropped V4 packet is excluded from the calculation.

Round 2 – LLQ + CBWFQ

The same 20 packets were reused. The six packets that got dropped as lower-priority traffic were F3, F1, W2, W3, W4, and F2. All six voice packets, all four video packets, and all three payment packets were delivered.

Voice average = (1 + 1 + 1 + 1 + 2 + 2) ÷ 6 = 1.33. Video average = (2 + 2 + 3 + 3) ÷ 4 = 2.50. Payments average = (3 + 3 + 4) ÷ 3 = 3.33.

Mission 6 Comparison

The result shows the most significant benefit of QoS implementation. The FIFO dropped one voice packet because it arrived late in the buffer, which was already full to 14. LLQ + CBWFQ protected all the voice, video, and payment packets by having a higher priority. The cost of this protection was a higher drop rate for the lower-priority FTP traffic and some web traffic. It is an acceptable cost to incur to maintain the fiesta's voice, video, and payment traffic. One limitation was that 4 packets per tick is an unrealistic simulation of a production network.

Tick		Packets delivered	
1		F3, VID3, W1, F1	
2		P3, VID1, V1, P2	
3		P1, VID2, V2, V3	
4		V5, V6	
Class	**Dropped**	**Average tick**	**Violation?**
Voice	1	3.20	YES — voice packet dropped
Video	1	2.00	—
Payments	0	2.33	—
Web	3	1.00	—
FTP	1	1.00	—

Tick	Packets delivered
1	V4, V5, V6, V1
2	V3, V2, VID3, VID1
3	VID2, VID4, P1, P2
4	P3, W1

Class	Dropped	Average tick	Violation?
Voice	0	1.33	NO
Video	0	2.50	NO
Payments	0	3.33	NO
Web	3	4.00	—
FTP	3	—	—

Traffic	FIFO drops	FIFO avg. tick	LLQ+CBWFQ drops	LLQ+CBWFQ avg. tick
Voice	1	3.20	0	1.33
Video	1	2.00	0	2.50
Payments	0	2.33	0	3.33
Web	3	1.00	3	4.00
FTP	1	1.00	3	—

Council Hearing — Defense Cheat Sheet

Question	Short answer
Why is the WAN the bottleneck?	The LAN is 100 Mbps but the WAN is only 2.048 Mbps; multiple traffic classes aggregate into the slower link (Slide 8).
Why LLQ for voice?	Voice is real-time and sensitive to delay, jitter, and loss; LLQ gives strict priority (Slides 13, 21).
Why not give everything priority?	Priority is limited; if too much traffic enters LLQ, it can starve other classes and reduce QoS effectiveness.
Why DiffServ?	It scales better across many routers and flows by using classes/marking instead of per-flow reservations (Slide 24).
What does WRED do?	It performs early congestion avoidance by dropping/marking packets before the queue is full, mainly helping TCP (Slides 31, 38, 39).
Shaping vs policing?	Shaping queues and smooths; policing drops/remarks excess. Shape outbound backup; ISP polices the CIR (Slide 40).
What happens to the 17th call?	17 calls require 1,088 kbps; voice+video+payments become 2,056 kbps, 8 kbps over. Use admission control.
What if staff marks FTP EF?	The trust boundary must re-mark untrusted EF so FTP cannot consume the strict-priority voice queue (Slide 37).